Brain Tumor

Overview

A brain tumor is an abnormal growth of cells within the brain or its surrounding structures. Brain tumors may be **benign (non-cancerous)** or **malignant (cancerous)**. Regardless of whether they are benign or malignant, brain tumors can interfere with normal brain function by compressing nearby brain tissue, increasing pressure inside the skull, or disrupting nerve pathways.

Brain tumors can occur at any age, although certain types are more common in children while others are more frequently seen in adults. Symptoms depend on the size, location, and growth rate of the tumor. Early diagnosis and appropriate treatment are important to improve outcomes and preserve neurological function.

Types of Brain Tumors

Brain tumors are broadly classified into **primary brain tumors**, which originate in the brain, and **secondary (metastatic) brain tumors**, which spread to the brain from cancers elsewhere in the body.

1. Primary Brain Tumors

Primary brain tumors develop directly from brain tissue or surrounding structures.

Common types include:

- Glioma
- Meningioma
- Pituitary adenoma
- Astrocytoma
- Oligodendroglioma
- Ependymoma

Some primary tumors are benign, while others are malignant.

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2. Secondary (Metastatic) Brain Tumors

Secondary brain tumors occur when cancer from another part of the body spreads to the brain.

Common primary cancers that spread to the brain include:

- Lung cancer
- Breast cancer
- Melanoma
- Kidney cancer
- Colon cancer

Metastatic brain tumors are more common than primary malignant brain tumors.

Causes

The exact cause of most brain tumors is unknown.

Possible contributing factors include:

- Genetic mutations
- Family history of certain genetic disorders
- Previous radiation exposure
- Rare inherited syndromes
- Increasing age for some tumor types

In most patients, no specific cause can be identified.

Risk Factors

Several factors may increase the risk of developing a brain tumor.

These include:

- Exposure to ionizing radiation
- Family history of brain tumors
- Certain inherited genetic conditions

- Increasing age
- Previous cancer in another organ

Although these factors increase risk, many individuals develop brain tumors without any known risk factors.

Symptoms

Symptoms depend on the tumor's size, location, and growth rate.

Common symptoms include:

- Persistent headache
- Headache that is worse in the morning
- Nausea
- Vomiting
- Seizures
- Blurred vision
- Double vision
- Hearing problems
- Difficulty speaking
- Difficulty walking
- Weakness in the arms or legs
- Loss of balance
- Memory problems
- Personality changes
- Difficulty concentrating
- Confusion
- Drowsiness

Symptoms usually become more noticeable as the tumor grows.

Neurological Changes

Patients may experience neurological symptoms depending on the affected part of the brain.

Common neurological changes include:

- Difficulty understanding speech
- Difficulty finding words
- Loss of coordination
- Reduced sensation
- Muscle weakness
- Facial weakness
- Vision loss
- Difficulty swallowing
- Tremors
- Changes in handwriting

These symptoms may develop gradually or, in some cases, suddenly.

Cognitive and Behavioral Symptoms

Brain tumors can affect thinking, emotions, and behavior.

Patients may develop:

- Memory loss
- Mood changes
- Irritability
- Depression
- Anxiety
- Personality changes

- Poor judgment
- Difficulty solving problems
- Reduced attention span
- Confusion

Family members often notice these changes before the patient does.

Diagnosis

Diagnosis involves identifying the tumor, determining its location, and evaluating whether it is benign or malignant.

Healthcare providers may perform:

- Detailed medical history
- Neurological examination
- Eye examination
- MRI of the brain
- CT scan
- Functional MRI (when appropriate)
- PET scan (selected cases)
- Biopsy
- Blood tests

MRI is usually the preferred imaging technique because it provides detailed images of brain structures.

A biopsy is often required to confirm the exact type of brain tumor.

Treatment

Treatment depends on the type, size, location, and grade of the tumor, as well as the patient's overall health.

Surgery

Whenever possible, surgery is performed to remove as much of the tumor as safely possible.

Surgery may:

- Relieve pressure on the brain
- Improve symptoms
- Provide tissue for diagnosis

Radiation Therapy

Radiation therapy uses high-energy beams to destroy tumor cells.

It may be used:

- After surgery
- Instead of surgery
- For tumors that cannot be completely removed

Chemotherapy

Chemotherapy uses medications to destroy cancer cells.

It may be administered:

- Orally
- Intravenously
- Directly into the surgical cavity in selected cases

Targeted Therapy

Certain brain tumors respond to targeted medications that attack specific molecular abnormalities within tumor cells.

These therapies are selected based on laboratory testing of the tumor tissue.

Rehabilitation

Patients often require rehabilitation after treatment.

Rehabilitation may include:

- Physical therapy
- Occupational therapy
- Speech therapy
- Cognitive rehabilitation
- Psychological counseling
- Nutritional support

The goal is to maximize independence and improve quality of life.

Possible Complications

Brain tumors and their treatment may lead to complications such as:

- Persistent seizures
- Permanent neurological deficits
- Vision problems
- Speech impairment
- Memory difficulties
- Hydrocephalus
- Brain swelling
- Weakness
- Balance problems
- Recurrence of the tumor

Regular follow-up helps identify complications early.

When to Seek Medical Attention

Immediate medical evaluation is recommended if a person develops:

- Persistent severe headache
- New-onset seizures
- Sudden weakness
- Difficulty speaking
- Vision loss
- Repeated vomiting
- Confusion
- Loss of consciousness
- Difficulty walking
- Personality changes
- Rapid worsening of neurological symptoms

Early diagnosis improves treatment options and outcomes.

Patient Advice

Patients should attend all follow-up appointments and take medications exactly as prescribed. Individuals receiving surgery, radiation therapy, or chemotherapy should report new neurological symptoms immediately. Participation in rehabilitation programs and maintaining a healthy lifestyle may improve recovery and overall quality of life. Emotional support from family members, healthcare professionals, and support groups is also an important part of long-term care.

Summary

A brain tumor is an abnormal growth of cells within the brain or surrounding tissues that may be benign or malignant. Symptoms vary depending on the tumor's location and size but commonly include persistent headaches, seizures, weakness, vision changes, memory problems, and personality changes. Diagnosis typically involves neurological examination, MRI, CT scan, and biopsy. Treatment may include surgery, radiation therapy, chemotherapy, targeted therapy, and rehabilitation. Early diagnosis and comprehensive treatment improve neurological function and quality of life.